

Degenerate Pivots for Shadow Rules on 0/1-Polytopes: A Corrected Antistalling Note

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Abstract

We revisit a proposed proof of a polynomial bound on degenerate pivots for the Ordered Shadow and Slim Shadow simplex rules of Black–De Loera–Kafer–Sanità. The proposed proof cannot be repaired as written: its central edge-realization step uses an auxiliary local objective, whereas a simplex pivot for the original linear program must be improving for the original objective. We give an explicit two-dimensional 0/1 example, written in standard equality form, where the auxiliary-objective step selects a degenerate basis direction with negative reduced cost for the original objective. Replacing the auxiliary objective by the original objective also does not force a prescribed shadow edge.

The positive result in this note is therefore partial. We prove a corrected edge-by-edge antistalling lemma for shadow edges under a precise support-compatibility hypothesis. Under that hypothesis, a fixed shadow edge can be realized by at most $\min\{N - M, M\} - 1$ degenerate pivots, all improving for the original objective, followed by the desired non-degenerate pivot. Consequently the Ordered Shadow and Slim Shadow geometric paths have polynomially many degenerate pivots under the same compatibility hypothesis along the run. We do not prove that this hypothesis follows from the pivot rules in Black–De Loera–Kafer–Sanità. Thus the result below is not a solution of their Question 2; it is a rigorous partial progress statement and an obstruction to the earlier attempted proof.

1 Status and relation to the open problem

Black–De Loera–Kafer–Sanità ask whether there is a polynomial bound on the number of degenerate pivots performed by the pivot rules they introduce for linear programs over 0/1-polytopes. Their paper proves linear bounds on the number of non-degenerate pivots for the Slim Shadow and Ordered Shadow rules, but deliberately leaves the degenerate-pivot count open.

The attempted proof which motivated this note tried to realize a prescribed shadow edge $[z, z^+]$ by choosing a local objective ℓ exposing that edge and then applying an antistalling argument with respect to ℓ . This is not a valid simplex implementation for the original problem unless every pivot selected by the ℓ -procedure is also improving for the original objective c . That compatibility is false in general, even for a square with one redundant inequality; see Proposition 3.1 below.

The main constructive statement here is Theorem 4.4. It gives a polynomial degenerate-pivot bound for one prescribed shadow edge, but only when the current bases satisfy a support condition that ensures the antistalling pivots are genuine c -improving simplex pivots and that the support of the desired edge decreases during stalling. Corollary 6.1 applies this edge theorem to the Ordered Shadow and Slim Shadow paths under the same hypothesis. Since that hypothesis is not currently derived from the rules of Black–De Loera–Kafer–Sanità, the corollary is conditional and should be read as partial progress, not as an answer to Question 2.

2 Simplex notation

We use standard equality form

$$\max\{c^T z : Az = b, z \geq 0\}, \quad (1)$$

where $A \in \mathbb{R}^{M \times N}$ has rank M . A basis is a set $B \subseteq [N]$ with $|B| = M$ and A_B nonsingular. Its nonbasic set is $N_B = [N] \setminus B$. The associated basic solution is

$$z_{N_B} = 0, \quad z_B = A_B^{-1}b.$$

A feasible basis has $z_B \geq 0$.

For $j \in N_B$, the simplex direction $p^j = p^j(B)$ is normalized by

$$p_j^j = 1, \quad p_k^j = 0 \quad (k \in N_B \setminus \{j\}), \quad Ap^j = 0. \quad (2)$$

Equivalently, $p_B^j = -A_B^{-1}A_j$. Its reduced cost for the maximization problem is

$$\bar{c}_j(B) = c^T p^j(B).$$

An entering index is c -improving if $\bar{c}_j(B) > 0$. The usual ratio test gives the maximal feasible step

$$\theta_j(B) = \min \left\{ \frac{z_i}{-p_i^j} : i \in B, p_i^j < 0 \right\},$$

with the convention that the minimum is $+\infty$ if the set is empty. A pivot is degenerate if $\theta_j(B) = 0$; then the basis changes but the represented point does not.

The basic cone at B is

$$C(B) = \text{cone}\{p^j(B) : j \in N_B\}.$$

Every feasible direction g at the represented point belongs to $C(B)$, and its conic coordinates in $C(B)$ are simply its nonbasic coordinates:

$$g = \sum_{j \in N_B} g_j p^j(B), \quad g_j \geq 0 \quad (j \in N_B). \quad (3)$$

We write

$$S_B(g) = \{j \in N_B : g_j > 0\}.$$

If $g = z^+ - z$ for another basic feasible solution z^+ , then

$$|S_B(g)| \leq \min\{N - M, M\}, \quad (4)$$

because $S_B(g) \subseteq N_B$ and, among nonbasic coordinates of z , the nonzero coordinates of g are nonzero coordinates of the basic feasible solution z^+ .

3 The local-objective edge-realization step is false

The next example isolates the exact failure. It also shows that merely replacing the auxiliary local objective by the original objective does not automatically force a prescribed edge.

Proposition 3.1. *There is a 0/1-polytope, a vertex z , an adjacent vertex z^+ , a feasible basis B representing z , an original objective c , and a local exposing objective ℓ such that $g = z^+ - z$ is c -improving and ℓ exposes the edge $[z, z^+]$, but the ℓ -guided antistalling step selects a degenerate simplex direction with negative reduced cost for c .*

Proof. Consider the standard equality form in variables

$$z = (x_1, x_2, s_1, s_2, s_3)$$

given by

$$x_1 + s_1 = 1, \quad x_2 + s_2 = 1, \quad x_1 + x_2 - s_3 = 0, \quad z \geq 0. \quad (5)$$

Projection to the x -coordinates gives the square $[0, 1]^2$; the variable s_3 is the slack for the redundant inequality $x_1 + x_2 \geq 0$.

Let

$$z^0 = (0, 0, 1, 1, 0), \quad z^+ = (0, 1, 1, 0, 1), \quad g = z^+ - z^0 = (0, 1, 0, -1, 1).$$

Use the feasible basis

$$B = \{s_1, s_2, x_1\}.$$

The two nonbasic variables are x_2 and s_3 . Directly from (2), the corresponding simplex directions are

$$p^{s_3} = (1, 0, -1, 0, 1), \quad \text{projecting to } (1, 0), \quad p^{x_2} = (-1, 1, 1, -1, 0), \quad \text{projecting to } (-1, 1).$$

Thus

$$g = p^{s_3} + p^{x_2}.$$

The projected feasible cone of the square at $(0, 0)$ is generated by $(1, 0)$ and $(0, 1)$, while the projected basic cone for B is generated by $(1, 0)$ and $(-1, 1)$. Hence p^{x_2} is not a feasible edge direction of the square. Indeed the ratio test is zero, because x_1 is a zero basic variable and $p_{x_1}^{x_2} = -1$.

Take the original objective

$$c = (2, 1, 0, 0, 0).$$

Then

$$c^T g = 1 > 0, \quad c^T p^{s_3} = 2 > 0, \quad c^T p^{x_2} = -1 < 0.$$

So g is improving for the original LP, but the degenerate basis direction p^{x_2} is not a valid improving simplex direction for the original objective.

Now take the local objective

$$\ell = (-2, 1, 0, 0, 0).$$

In the projected square, ℓ is positive on the edge direction $(0, 1)$ and negative on the other feasible edge direction $(1, 0)$; hence it exposes the oriented edge from $(0, 0)$ to $(0, 1)$. However

$$\ell^T p^{s_3} = -2 < 0, \quad \ell^T p^{x_2} = 3 > 0.$$

Therefore an antistalling rule driven by the exposing objective ℓ selects the degenerate direction p^{x_2} , but that direction has negative reduced cost for the original objective c . It is not a permissible simplex pivot for the original linear program. $\square$

Remark 3.2. *The example also shows why the naive repair “use c instead of ℓ ” does not realize an arbitrary prescribed edge. With the same basis and prescribed direction $g = (0, 1, 0, -1, 1)$, the only c -improving direction in the support of g is p^{s_3} , which is a non-degenerate pivot to the other square vertex $(1, 0)$ rather than to $(0, 1)$. Thus an original-objective antistalling theorem by itself bounds stalling before some improving edge pivot; it does not force the chosen shadow edge.*

4 A corrected edge-by-edge antistalling statement

We now give a valid replacement for the false edge-realization lemma. The statement is conditional. It is designed to make explicit the compatibility that the failed proof implicitly assumed.

Let z be a vertex of (1), and let $g = z^+ - z$ be the direction from z to an adjacent vertex z^+ . Let v be the auxiliary vector defining a shadow projection, and assume

$$v^T g > 0, \quad c^T g > 0.$$

Set

$$\rho = \frac{c^T g}{v^T g}, \quad a = c - \rho v. \quad (6)$$

Thus $a^T g = 0$. In the shadow interpretation, a is the linear functional whose zero set supports the projected edge of slope ρ .

Definition 4.1 (shadow-exposed edge). *The oriented edge direction g is called shadow-exposed at z by (c, v) if*

$$a^T h \leq 0 \quad (7)$$

for every feasible direction h at z , where $a = c - \rho v$ is defined by (6), and equality in (7) holds on feasible edge directions only for positive multiples of g .

For a non-tied edge on the upper path of a two-dimensional shadow, this is exactly the usual supporting-line condition.

Definition 4.2 (support-compatible basis for g). *Let B be a feasible basis representing z . We say that B is support-compatible with g and v if*

$$v^T p^j(B) > 0 \quad \text{for every } j \in S_B(g). \quad (8)$$

We say that B is non-tied on the support of g if

$$a^T p^j(B) = 0 \text{ and } j \in S_B(g) \implies p^j(B) \text{ is a positive multiple of } g. \quad (9)$$

Lemma 4.3 (one support-decreasing degenerate pivot). *Assume g is shadow-exposed at z by (c, v) . Let B be a feasible basis representing z and satisfying (8). If $j \in S_B(g)$ and*

$$a^T p^j(B) > 0,$$

then $p^j(B)$ is a c -improving degenerate simplex direction. Moreover, there exists a zero basic index $i \in B$ such that

$$z_i = 0, \quad g_i = 0, \quad p_i^j(B) < 0. \quad (10)$$

Pivoting with entering index j and such a leaving index i is degenerate and strictly decreases $|S_B(g)|$.

Proof. Since $j \in S_B(g)$ and B is support-compatible, $v^T p^j(B) > 0$. Hence

$$c^T p^j(B) = a^T p^j(B) + \rho v^T p^j(B) > 0,$$

so the direction is c -improving.

The direction $p^j(B)$ cannot be feasible at z . If it were feasible, then (7) would give $a^T p^j(B) \leq 0$, contradicting the assumption $a^T p^j(B) > 0$. Therefore some zero coordinate of z must become

negative along $p^j(B)$; equivalently, there is a zero basic index $i \in B$ with $p_i^j(B) < 0$. This already shows that the ratio test for entering j has value zero.

It remains to prove that one can choose such an i with $g_i = 0$. Suppose not. Then every zero coordinate made negative by $p^j(B)$ is positive in the direction g . For sufficiently small $\varepsilon > 0$, the vector

$$g + \varepsilon p^j(B)$$

is still a feasible direction at z : coordinates with $g_i > 0$ remain nonnegative for small ε , and coordinates with $g_i = 0$ are not made negative by the supposition. But

$$a^T(g + \varepsilon p^j(B)) = a^T g + \varepsilon a^T p^j(B) = \varepsilon a^T p^j(B) > 0,$$

contradicting the shadow-exposed condition (7). Thus an index i satisfying (10) exists.

After the pivot, the represented point is still z because $z_i = 0$. The old entering index j becomes basic, so it leaves the nonbasic support of g . The old leaving index i becomes nonbasic, but $g_i = 0$, so it does not enter the nonbasic support. All other nonbasic coordinates of g are unchanged. Therefore $|S_B(g)|$ decreases by one. $\square$

Theorem 4.4 (conditional realization of one shadow edge). *Let $g = z^+ - z$ be a shadow-exposed adjacent edge direction at z . Start from a feasible basis B_0 representing z . Consider the following fixed-edge antistalling rule while the represented point remains z :*

- (a) *If there is $j \in S_B(g)$ with $a^T p^j(B) > 0$, choose one such j as entering index and choose a leaving index i satisfying (10).*
- (b) *If no such j exists, choose an index $j \in S_B(g)$ with $a^T p^j(B) = 0$.*

Assume that every basis encountered before the first non-degenerate pivot is support-compatible with g and v , and is non-tied on the support of g in the sense of Definition 4.2. Then the rule performs at most

$$|S_{B_0}(g)| - 1 \leq \min\{N - M, M\} - 1 \tag{11}$$

degenerate pivots, all improving for the original objective c , followed by one non-degenerate pivot from z to z^+ .

Proof. At any encountered basis B , expand g as in (3):

$$g = \sum_{j \in S_B(g)} g_j p^j(B), \quad g_j > 0.$$

Because $a^T g = 0$, we have

$$\sum_{j \in S_B(g)} g_j a^T p^j(B) = 0. \tag{12}$$

If some term in (12) is positive, the rule uses case (a). Lemma 4.3 shows that the chosen pivot is a valid c -improving degenerate simplex pivot and that $|S_B(g)|$ decreases by one.

Since the support size is a nonnegative integer, case (a) can occur at most $|S_{B_0}(g)| - 1$ times before either a non-degenerate pivot occurs or no positive term remains in (12). If no positive term remains, then (12) forces

$$a^T p^j(B) = 0 \quad \text{for all } j \in S_B(g).$$

By the non-tie assumption, each such $p^j(B)$ is a positive multiple of g . Since the simplex directions $p^j(B)$ have distinct unit nonbasic coordinates, this can happen for only one index $j \in S_B(g)$.

Pivoting on that index moves along the edge ray generated by g . Because z^+ is the adjacent endpoint of that ray, the non-degenerate pivot reaches z^+ .

Finally, the support bound (11) follows from (4). $\square$

Remark 4.5. *Theorem 4.4 is deliberately stated with explicit hypotheses. The failed proof effectively assumed that the exposing functional could be used to choose entering indices. The corrected theorem instead chooses only directions that are improving for the original objective c . The price is the support-compatibility hypothesis (8). I do not know how to derive that hypothesis from the Ordered Shadow or Slim Shadow pivot rules in full generality.*

5 A simpler unconditional partial case

The following consequence of the antistalling theorem of Kukhareno–Sanità is weaker than Theorem 4.4, but it is unconditional in a transparent special case.

Proposition 5.1 (edge-forced vertices). *Suppose z is a nonoptimal vertex of (1) and has exactly one c -improving incident edge, namely the edge from z to z^+ . Then, from any feasible basis representing z , the antistalling rule of Kukhareno–Sanità guided by $g = z^+ - z$ performs at most*

$$\min\{N - M, M\} - 1$$

degenerate pivots before making the non-degenerate pivot from z to z^+ .

Proof. The direction $g = z^+ - z$ is a feasible c -improving direction of the form “basic feasible solution minus current basic feasible solution.” The theorem of Kukhareno–Sanità bounds the number of consecutive degenerate pivots, when guided by such a direction, by the initial nonbasic support size minus one; by (4), this is at most $\min\{N - M, M\} - 1$. The first non-degenerate pivot produced by their rule is a c -improving simplex pivot, hence it moves from z along a c -improving incident edge. By uniqueness, that edge is $[z, z^+]$. $\square$

Corollary 5.2. *Let*

$$z^0, z^1, \dots, z^L$$

be a c -monotone path of vertices such that, for each $k < L$, the edge $[z^k, z^{k+1}]$ is the unique c -improving edge incident to z^k . Then the path can be implemented by genuine simplex pivots with at most

$$L(\min\{N - M, M\} - 1)$$

degenerate pivots before reaching z^L .

This corollary is not meant to describe generic shadow-rule behavior; shadow vertices usually have many improving incident edges. Its purpose is to record a clean case where the Kukhareno–Sanità theorem really does turn a known geometric path into a polynomially stalled simplex path.

6 Conditional consequence for the shadow rules

Let $P \subseteq [0, 1]^p$ be the original 0/1-polytope, and let (1) be a standard equality form obtained from its inequality description by adding slacks. This standard form is affinely isomorphic to P , so vertices and edges correspond. Let

$$z^0, z^1, \dots, z^L$$

be the vertex path selected by one of the geometric shadow rules of Black–De Loera–Kafer–Sanità. For Slim Shadow one has $L \leq p$, and for Ordered Shadow one has $L \leq d = \dim(P)$.

Corollary 6.1 (conditional shadow bound). *Assume that for every $k < L$, the edge direction*

$$g^k = z^{k+1} - z^k$$

is shadow-exposed at z^k by the corresponding pair (c, v^k) , and that every basis encountered by the fixed-edge antistalling rule of Theorem 4.4 is support-compatible and non-tied on the support of g^k . Then the shadow path can be implemented by genuine simplex pivots with at most

$$L(\min\{N - M, M\} - 1) \tag{13}$$

degenerate pivots before the final vertex is reached. In particular, under these hypotheses,

$$\text{Slim Shadow: } p(\min\{N - M, M\} - 1), \quad \text{Ordered Shadow: } d(\min\{N - M, M\} - 1)$$

degenerate pivots suffice before reaching an optimal vertex.

Proof. Apply Theorem 4.4 independently to each edge of the shadow path. For the k th edge, the number of degenerate pivots is at most

$$|S_{B_k}(g^k)| - 1 \leq \min\{N - M, M\} - 1,$$

where B_k is the basis present when the algorithm first represents z^k . Summing over the L shadow edges gives (13). The bounds for Slim Shadow and Ordered Shadow follow by substituting the known non-degenerate edge bounds $L \leq p$ and $L \leq d$, respectively. $\square$

Remark 6.2 (terminal optimal bases). *If one insists on ending at an optimal basis rather than merely an optimal vertex, the terminal cleanup lemma of Kukhareno–Sanità gives an additional bound of at most $N - M$ degenerate pivots at the final vertex. This term is independent of the number of shadow edges. The issue left open here is not terminal cleanup, but the support compatibility required before and during the edge realizations.*

7 What remains open

The attempted proof is not a solution of Question 2 because it changes the objective used to select simplex pivots. Proposition 3.1 shows this failure on a 0/1 square with one redundant constraint. Remark 3.2 shows that using the original objective alone still does not force a prescribed edge.

The valid progress is the following.

- (i) The obstruction is now explicit: to realize a prescribed shadow edge, one must control not only feasibility of the desired edge but also the reduced costs of all degenerate basis directions used before the edge appears.
- (ii) Theorem 4.4 gives a polynomial bound for one edge when the support-compatibility and no-tie hypotheses hold.
- (iii) Corollary 6.1 converts the linear non-degenerate shadow bounds into polynomial total degenerate bounds under the same hypotheses along the run.
- (iv) Proposition 5.1 gives an unconditional polynomial antistalling statement for edge-forced vertices.

The missing step needed for a solution of Black–De Loera–Kafer–Sanità’s Question 2 is to prove that their actual basis-level pivot rules, including the degenerate phase used to obtain shadow-compatible bases, satisfy a condition strong enough to imply Theorem 4.4, or to find a different potential that bounds that degenerate phase directly. This note does not prove either statement.

References

- [1] Alexander E. Black, Jesús A. De Loera, Sean Kafer, and Laura Sanità. On the Simplex method for 0/1 polytopes. *arXiv:2111.14050*, 2021.
- [2] Kirill Kukharenko and Laura Sanità. On the number of degenerate simplex pivots. *arXiv:2311.15799*, version 4, 2026.